

ASSINGMENT ON CSE - 423

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Answer to the Question No. 1

The line segment from $(5, -15)$ to $(0, -9)$

Using DDA Algorithm

$$\Delta y = -9 - -15 = 10$$

$$\Delta x = 0 - 5 = -5 \Rightarrow -5$$

$$\text{So, slope} \rightarrow \frac{\Delta y}{\Delta x} = \frac{10}{-5} = -2$$

We can see that,

slope $-1 < m < 1$ is not in range.

$$\text{As, } m = -2 \rightarrow 1/m = -0.5$$

$y(+1)$	$x (+1/m)$	round of x	pixel
-15	5	5	$(5, -15)$
-14	4.5	5	$(5, -14)$
-13	4	4	$(4, -13)$
-12	3.5	4	$(4, -12)$
-11	3	3	$(3, -11)$
-10	2.5	3	$(3, -10)$
-9	2	2	$(2, -9)$

(b)

We are to compute the find fixel using
8-way symmetry $(5, -15)$ to $(0, -5)$

$$dx = -5$$

The zone will not be conventional

$$dy = 10$$

It would either be 2 or 3

$$m = -2$$

$|dy| > dx \rightarrow$ closer to y-axis

So, it is zone -2

For zone -2 $\rightarrow$ zone 0, $x = y$, $y = -x$

$(-15, -5)$ and $(-5, 0)$

$$dx = 10, dy = 5$$

$$d_{init} = 2(5) - 10 = 0$$

$$\Delta NE = -10, \Delta E = 10$$

x	y	d	NE/E	d_{update}	$x = -y$	$y = x$
5	15	0	E	10	-15	5
6	14	10	NE	20	-14	6
7	13	20	NE	30	-13	7
8	12	30	NE	40	-12	8
9	11	40	NE	50	-11	9
10	10	50	NE	60	-10	10
11	9	60	NE	70	-9	11
12	8	70	NE	80	-8	12
13	7	80	NE	90	-7	13
14	6	90	NE	100	-6	14
15	5	100	NE	110	-5	15

Answer to the Question No. 2

(a)

We use bitwise AND Operation during Outcode of Cohen-Sutherland Algorithm.

In Cohen-Sutherland algorithm we know there is a 4 bit region code and we have made to clipping decisions. Through using AND Operation on Outcodes we can identify which ones are completely accepted, completely rejected and partially accepted. Thus it helps in identifying the lines which are not required in calculations and unnecessary clipping can be avoided.

(b)

We can see $x_0, y_0 \rightarrow (1, 3)$

The standard parametric is $P(t) = (x_0, y_0) + t(\bar{D})$

$$\bar{D} = (6, 7), \quad t = 0.3$$

So, the co-ordinates using parametric equation is :- $(1, 3) + 0.3(6, 7) \rightarrow (2.8, 5.1)$

(i) Cohen - Sutherland Algorithm

So, for the clipping window

$$x_{\min} = -50, \quad y_{\min} = -10, \quad x_{\max} = 60, \quad y_{\max} = 100$$

$$x_1 = -50, \quad y_1 = -70, \quad x_2 = 40, \quad y_2 = 100 \quad (P_1, L P_2)$$

For the region codes :-

TRRL

$$P_1 \rightarrow 0100$$

$$P_2 \rightarrow 0000$$

So, conducting Bitwise AND Operation

$$\begin{array}{r} 0100 \\ 0000 \\ \hline 0000 \end{array}$$

So, P_1 AND P_2 we get 0000. Thus it is partially inside.

(ii) Cyrus - Beck Algorithm

$$x_{\min} = -50, \quad y_{\min} = -10, \quad x_{\max} = 60, \quad y_{\max} = 100$$

$$x_1 = -50, \quad y_1 = -70, \quad x_2 = 40, \quad y_2 = 100$$

$$\vec{D} = (40, -50, 100, -70)$$

$$= (90, 170)$$

Boundary	N_i	$N_i \cdot D$	t	PE / PL	t_E	t_L
Left	$(-1, 0)$	-90	0	PE	0	1
Right	$(1, 0)$	90	1.22	PL	0	1
Bottom	$(0, -1)$	-170	0.35	PE	0.35	1
Top	$(0, 1)$	170	1	PL	0.35	1

So, the 4 values of t has been calculated using Cyrus - Beck Algorithm